SETK 4643

Thermodynamics

Chapter 4: Energy Analysis Of Closed Systems

Dr. Nur Hidayah Zainan

4.1 Moving Boundary Work

- One form of mechanical work frequently encountered in practice is associated with the expansion or compression of a gas in a piston-cylinder device.
- During this process, part of the boundary (the inner face of the piston) moves back and forth.
- Therefore, the expansion and compression work is often called moving boundary work, or simply **boundary work**, sometimes also called ***PdV* work**

- The moving boundary work associated with real engines or compressors cannot be determined exactly from a thermodynamic analysis alone
  - the piston usually moves at very high speeds, making it difficult for the gas inside to maintain equilibrium.
  - Thus, the states through which the system passes during the process cannot be specified, and no process path can be drawn.
  - Work, being a path function, cannot be determined analytically without a knowledge of the path.
  - Therefore, the boundary work in real engines or compressors is determined by direct measurements.

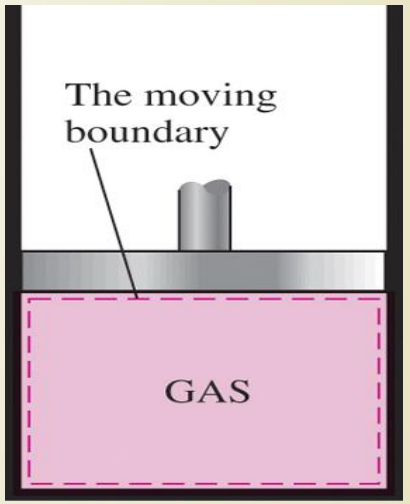


Fig. 4.1 The work associated with a moving boundary is called boundary work.

4.1 Moving Boundary Work (2)

- If the piston is allowed to move a distance *ds* in a quasi-equilibrium manner, the differential work done during this process is $\delta W_b = F\,ds = PA\,ds = P\,dV$
- *P* is the absolute pressure, which is always positive.
- However, the volume change *dV* is
  - positive during an expansion process (volume Increasing), and
  - negative during a compression process (volume decreasing).
- Thus, the boundary work is:
  - positive during an expansion process, and
  - negative during compression process

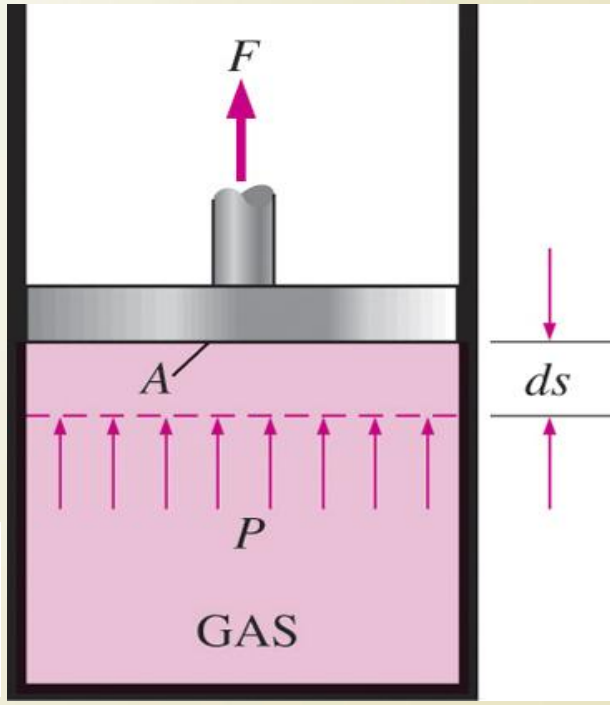


Fig. 4.2 A gas does a differential amount of work δW_b as it forces the piston to move by a differential amount *ds*.

4.1 Moving Boundary Work (3)

- The total boundary work done during the entire process as the piston moves is by adding all the differential works from the initial state to the final state:

$$W_b = \int_1^2 PdV \text{ (kJ) ----- (4.2)}$$

- As in the diagram, the differential area *dA* is equal to *PdV*, which is the differential work.
- The total area *A* under the process curve 1-2 is obtained by adding theses differential areas:

$$Area = A = \int_1^2 dA = \int_1^2 PdV = W_b$$

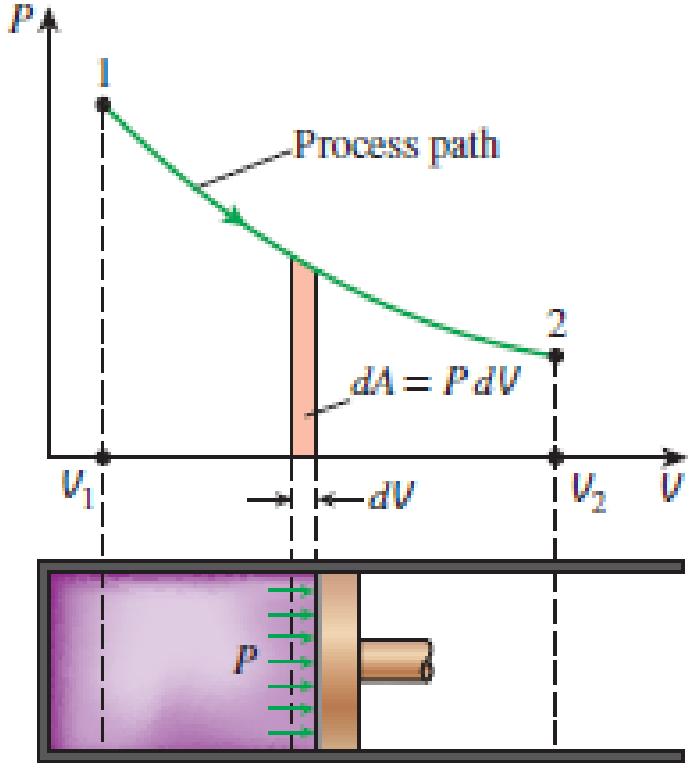


FIGURE 4–3

The area under the process curve on a *P-V* diagram represents the boundary work.

4.1 Moving Boundary Work (4)

- A gas can follow several different paths as it expands from state 1 to state 2.
- In general, each path will have a different area underneath it.
  - Since this area represents the magnitude of the work, the work done will be different for each process
  - This is expected, since work is a path function.
  - If work were not a path function, no cyclic devices (car engines, power plants) could operate as work-producing devices.
    - The work produced by cyclic devices during one part of the cycle would have to be consumed during another part, and there would be no **net work output**.

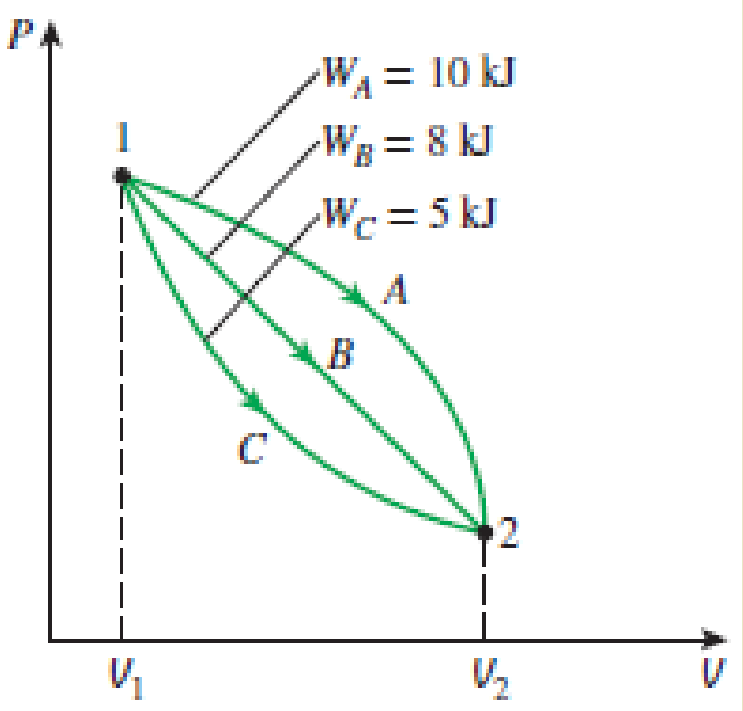


FIGURE 4–4
The boundary work done during a process depends on the path followed as well as the end states.

4.1 Moving Boundary Work (5)

- The cycle shown in Fig. 4.5 produces a net work output
  - Since the work done by the system during the **expansion process** (area under path *A*) is greater than the work done on the system during the **compression part** of the cycle (area under path *B*), and
  - The difference between these two is the **net work done during the cycle**.

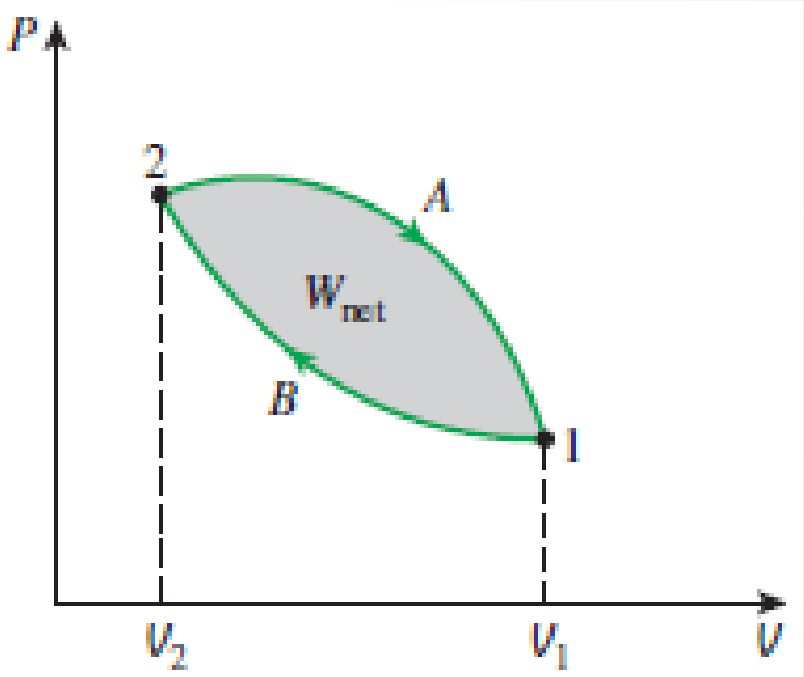


FIGURE 4–5
The net work done during a cycle is the difference between the work done by the system and the work done on the system.

EXAMPLE 4–1 Boundary Work for a Constant-Volume Process

A rigid tank contains air at 500 kPa and 150°C. As a result of heat transfer to the surroundings, the temperature and pressure inside the tank drop to 65°C and 400 kPa, respectively. Determine the boundary work done during this process.

SOLUTION Air in a rigid tank is cooled, and both the pressure and temperature drop. The boundary work done is to be determined.

Analysis A sketch of the system and the P-V diagram of the process are shown in Fig. 4–6. The boundary work can be determined from Eq. 4–2 to be

$$W_b = \int_1^2 P\, \cancelto{0}{dV} = 0$$

Discussion This is expected since a rigid tank has a constant volume and $dV = 0$ in this equation. Therefore, there is no boundary work done during this process. That is, the boundary work done during a constant-volume process is always zero. This is also evident from the P-V diagram of the process (the area under the process curve is zero).

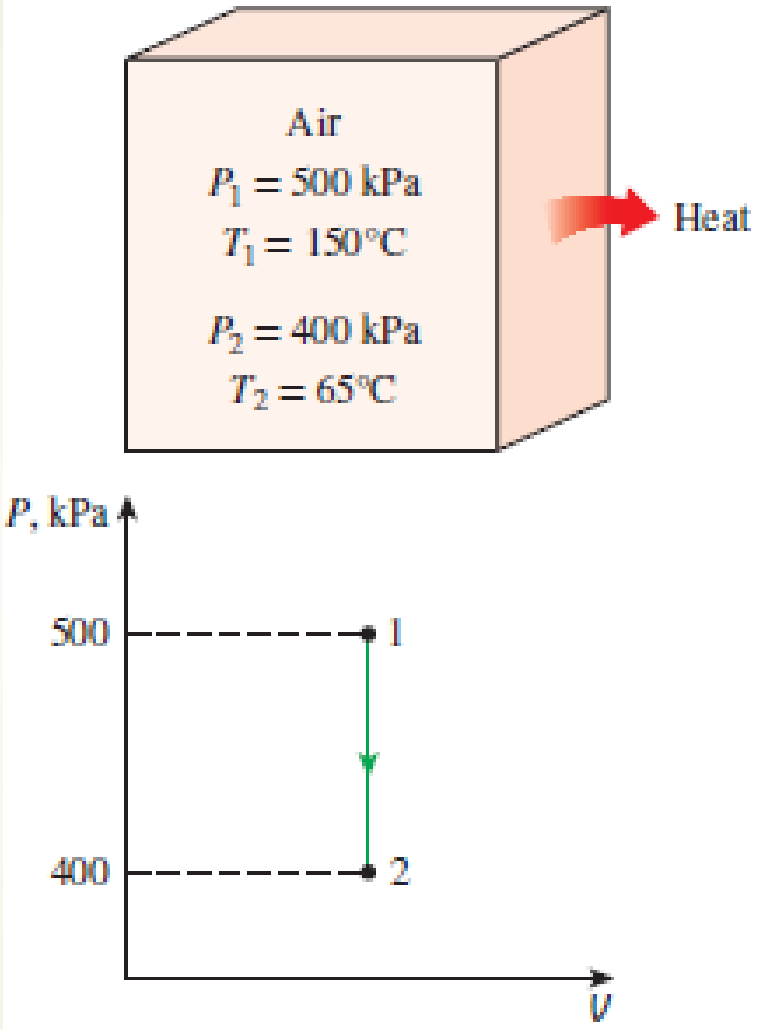


FIGURE 4–6
Schematic and P-V diagram for Example 4–1.

EXAMPLE 4–2 Boundary Work for a Constant-Pressure Process

A frictionless piston–cylinder device contains 10 lbm of steam at 60 psia and 320°F. Heat is now transferred to the steam until the temperature reaches 400°F. If the piston is not attached to a shaft and its mass is constant, determine the work done by the steam during this process.

SOLUTION Steam in a piston–cylinder device is heated and the temperature rises at constant pressure. The boundary work done is to be determined.

Assumptions The expansion process is quasi-equilibrium.

Analysis A sketch of the system and the P-v diagram of the process are shown in Fig. 4–7. Even though it is not explicitly stated, the pressure of the steam within the cylinder remains constant during this process since both the atmospheric pressure and the weight of the piston remain constant. Therefore, this is a constant-pressure process, and, from Eq. 4–2

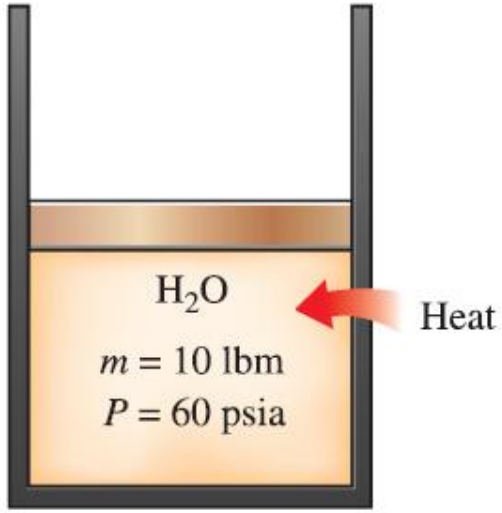
$$W_b = \int_1^2 P\,dV = P_0 \int_1^2 dV = P_0(V_2 - V_1) \qquad \textbf{(4–6)}$$

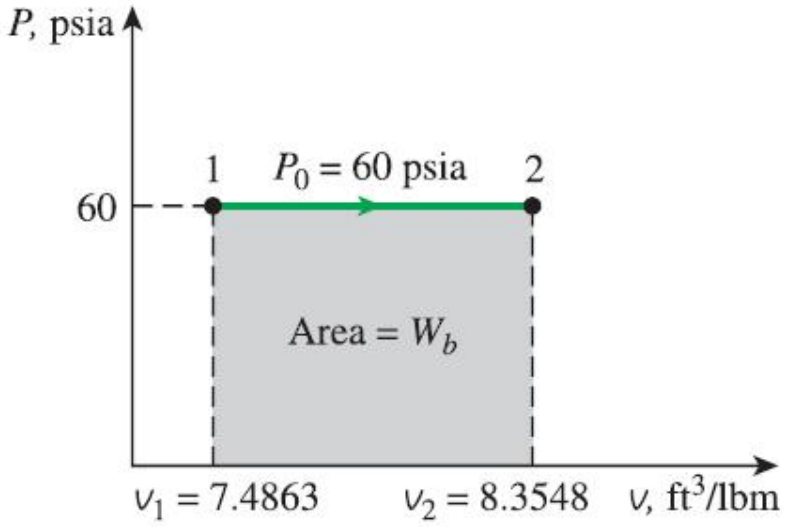
or

$$W_b = mP_0(v_2 - v_1)$$

since $V = mv$. From the superheated vapor table (Table A–6E), the specific volumes are determined to be $v_1 = 7.4863$ ft³/lbm at state 1 (60 psia, 320°F) and $v_2 = 8.3548$ ft³/lbm at state 2 (60 psia, 400°F). Substituting these values yields

$$W_b = (10\text{ lbm})(60\text{ psia})[(8.3548 - 7.4863)\text{ft}^3/\text{lbm}]\left(\frac{1\text{ Btu}}{5.404\text{ psia}\cdot\text{ft}^3}\right)$$
$$= \mathbf{96.4\ Btu}$$





Redo the problem for:

$P = 400$ kPa, $m = 5$ kg, $T_1 = 200$ °C, $T_2 = 300$ °C

EXAMPLE 4–3 Isothermal Compression of an Ideal Gas

A piston–cylinder device initially contains 0.4 m^3 of air at 100 kPa and 80°C. The air is now compressed to 0.1 m^3 in such a way that the temperature inside the cylinder remains constant. Determine the work done during this process.

SOLUTION Air in a piston–cylinder device is compressed isothermally. The boundary work done is to be determined.

Assumptions **1** The compression process is quasi-equilibrium. **2** At specified conditions, air can be considered to be an ideal gas since it is at a high temperature and low pressure relative to its critical-point values.

Analysis A sketch of the system and the P-V diagram of the process are shown in Fig. 4–8. For an ideal gas at constant temperature T_0,

$$PV = mRT_0 = C \quad \text{or} \quad P = \frac{C}{V}$$

where C is a constant. Substituting this into Eq. 4–2, we have

$$W_b = \int_1^2 P\,dV = \int_1^2 \frac{C}{V}dV = C\int_1^2 \frac{dV}{V} = C \ln\frac{V_2}{V_1} = P_1V_1 \ln\frac{V_2}{V_1} \qquad \textbf{(4–7)}$$

In Eq. 4–7, P_1V_1 can be replaced by P_2V_2 or mRT_0. Also, V_2/V_1 can be replaced by P_1/P_2 for this case since $P_1V_1 = P_2V_2$.

Substituting the numerical values into Eq. 4–7 yields

$$W_b = (100\text{ kPa})(0.4\text{ m}^3)\left(\ln\frac{0.1}{0.4}\right)\left(\frac{1\text{ kJ}}{1\text{ kPa·m}^3}\right)$$

$$= \mathbf{-55.5\text{ kJ}}$$

Discussion The negative sign indicates that this work is done on the system (a work input), which is always the case for compression processes.

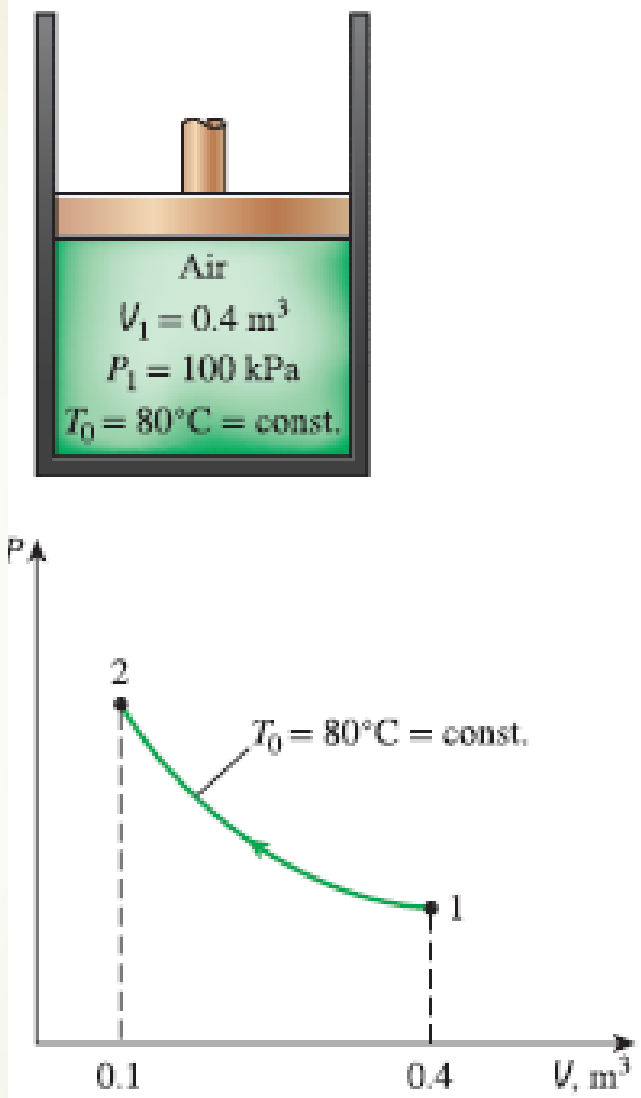
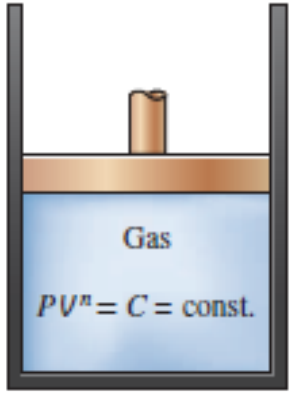


FIGURE 4–8
Schematic and P-V diagram for Example 4–3.

4.1 Moving Boundary Work (6)

Polytropic Process

- During actual expansion and compression processes of gases, pressure and volume are often related by $PV^n = C$, where n and C are constants.
- A process of this kind is called a polytropic process (Fig. 4.6).



- Work done during a polytropic process:
  - The pressure for a polytropic process can be expressed as

$$P = CV^{-n}$$

  - Substituting this relation into Eq. 4.2,

$$W_b = \int_1^2 P\,dV = \int_1^2 CV^{-n}\,dV = C\frac{V_2^{-n+1} - V_1^{-n+1}}{-n+1} = \frac{P_2V_2 - P_1V_1}{1-n}$$

Since $C = P_1V_1^n = P_2V_2^n$

  - For Ideal gas (*PV=mRT*): $W_b = \dfrac{mR(T_2 - T_1)}{1-n}$

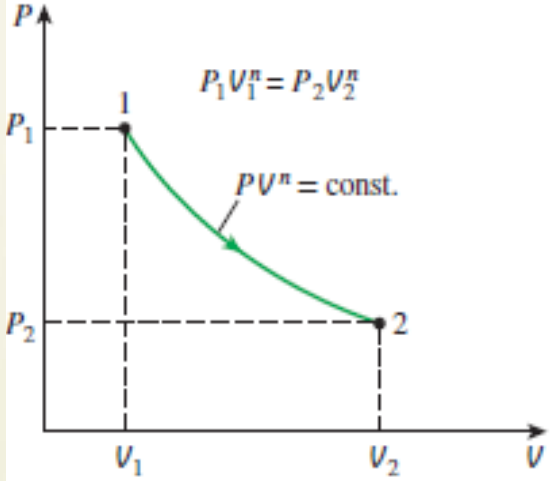


FIGURE 4–9
Schematic and P-V diagram for a polytropic process.

4.2 Energy Balance for Closed Systems

- The energy balance for any system undergoing any process can be expressed as (see Chapter 2):

$$\underbrace{E_{in} - E_{out}}_{\substack{\text{Net energy transfer} \\ \text{by heat, work, and mass}}} = \underbrace{\Delta E_{system}}_{\substack{\text{Change in internal, kinetic,} \\ \text{potential, etc., energies}}} \qquad (kJ)$$

$$E_{\text{in}} - E_{\text{out}} = (Q_{\text{in}} - Q_{\text{out}}) + (W_{\text{in}} - W_{\text{out}}) + (E_{\text{mass, in}} - E_{\text{mass, out}}) = \Delta E_{\text{system}}$$

- Or, in the rate form, as

$$\underbrace{\dot{E}_{in} - \dot{E}_{out}}_{\substack{\textit{Rate of net energy transfer} \\ \textit{by heat, work, and mass}}} = \underbrace{dE_{system}/dt}_{\substack{\textit{Rate of change in internal, kinetic,} \\ \textit{potential, etc., energies}}} = 0 \qquad (\text{kW})$$

4.2 Energy Balance for Closed Systems (2)

- For a **closed system undergoing a cycle**, the initial and final states are identical, thus $\Delta E_{system} = E_2 - E_1 = 0$.
  - The energy balance for a cycle simplifies to $E_{in} - E_{out} = 0$ or $E_{in} = E_{out}$
  - It can also be expressed in terms of heat and work interactions as

$$W_{net,out} = Q_{net,in} \quad \text{or} \quad \dot{W}_{net,out} = \dot{Q}_{net,in} \quad \text{(for a cycle)}$$

*the **net work output** during a cycle is equal to **net heat input***

- Using the classical thermodynamics sign convention and assuming:
  - heat to be transferred into the system (heat input) in the amount of Q
  - work to be done by the system (work output) in the amount of W:

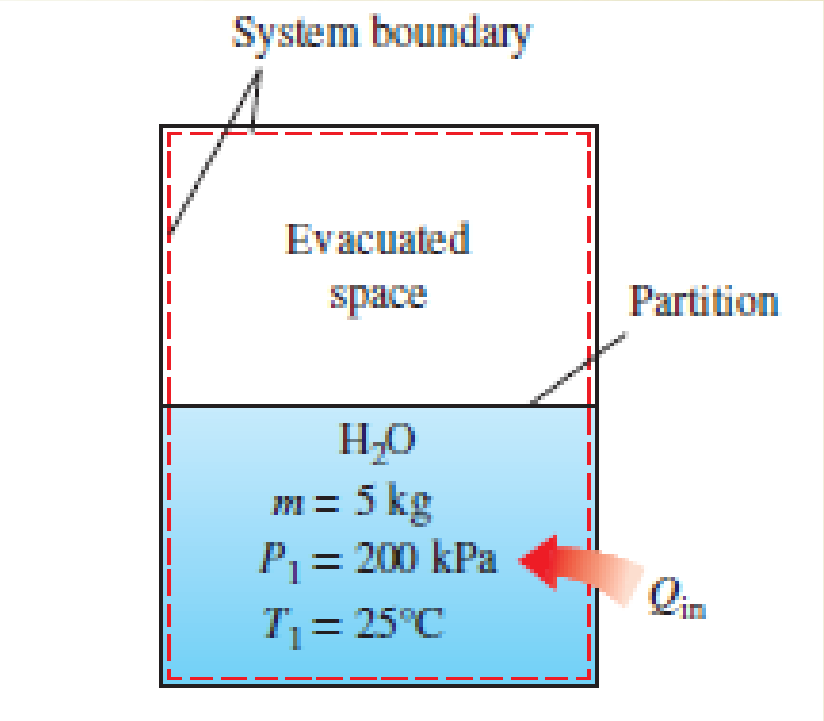
$$Q_{net,in} - W_{net,out} = \Delta E_{system} \quad \text{or} \quad Q - W = \Delta E$$

where $Q = Q_{net,in} = Q_{in} - Q_{out}$ is the net heat input

$W = W_{net,out} = W_{out} - W_{in}$ is the net work output.

EXAMPLE 4–6 Unrestrained Expansion of Water

A rigid tank is divided into two equal parts by a partition. Initially, one side of the tank contains 5 kg of water at 200 kPa and 25°C, and the other side is evacuated. The partition is then removed, and the water expands into the entire tank. The water is allowed to exchange heat with its surroundings until the temperature in the tank returns to the initial value of 25°C. Determine (*a*) the volume of the tank, (*b*) the final pressure, and (*c*) the heat transfer for this process.



SOLUTION One-half of a rigid tank is filled with liquid water while the other side is evacuated. The partition between the two parts is removed, and water is allowed to expand and fill the entire tank while the temperature is maintained constant. The volume of the tank, the final pressure, and the heat transfer are to be to determined.

Assumptions **1** The system is stationary and thus the kinetic and potential energy changes are zero, $\Delta KE = \Delta PE = 0$ and $\Delta E = \Delta U$. **2** The direction of heat transfer is to the system (heat gain, Q_{in}). A negative result for Q_{in} indicates the assumed direction is wrong and thus it is a heat loss. **3** The volume of the rigid tank is constant, and thus there is no energy transfer as boundary work. **4** There is no electrical, shaft, or any other kind of work involved.

Analysis We take the contents of the tank, including the evacuated space, as the *system* (Fig. 4–15). This is a *closed system* since no mass crosses the system boundary during the process. We observe that the water fills the entire tank when the partition is removed (possibly as a liquid–vapor mixture).

(*a*) Initially the water in the tank exists as a compressed liquid since its pressure (200 kPa) is greater than the saturation pressure at 25°C (3.1698 kPa). Approximating the compressed liquid as a saturated liquid at the given temperature, we find

$$v_1 \cong v_{f\,@\,25°C} = 0.001003 \text{ m}^3/\text{kg} \cong 0.001 \text{ m}^3/\text{kg} \quad \text{(Table A–4)}$$

Then the initial volume of the water is

$$V_1 = mv_1 = (5 \text{ kg})(0.001 \text{ m}^3/\text{kg}) = 0.005 \text{ m}^3$$

The total volume of the tank is twice this amount:

$$V_{\text{tank}} = (2)(0.005 \text{ m}^3) = \mathbf{0.01 \text{ m}^3}$$

(*b*) At the final state, the specific volume of the water is

$$v_2 = \frac{V_2}{m} = \frac{0.01 \text{ m}^3}{5 \text{ kg}} = 0.002 \text{ m}^3/\text{kg}$$

which is twice the initial value of the specific volume. This result is expected since the volume doubles while the amount of mass remains constant.

$$\text{At } 25°\text{C:} \quad v_f = 0.001003 \text{ m}^3/\text{kg} \quad \text{and} \quad v_g = 43.340 \text{ m}^3/\text{kg} \quad \text{(Table A–4)}$$

Since $v_f < v_2 < v_g$, the water is a saturated liquid–vapor mixture at the final state, and thus the pressure is the saturation pressure at 25°C:

$$P_2 = P_{\text{sat}\,@\,25°\text{C}} = \mathbf{3.1698 \text{ kPa}} \quad \text{(Table A–4)}$$

(*c*) Under stated assumptions and observations, the energy balance on the system can be expressed as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\substack{\text{Net energy transfer}\\ \text{by heat, work, and mass}}} = \underbrace{\Delta E_{\text{system}}}_{\substack{\text{Change in internal, kinetic,}\\ \text{potential, etc., energies}}}$$

$$Q_{\text{in}} = \Delta U = m(u_2 - u_1)$$

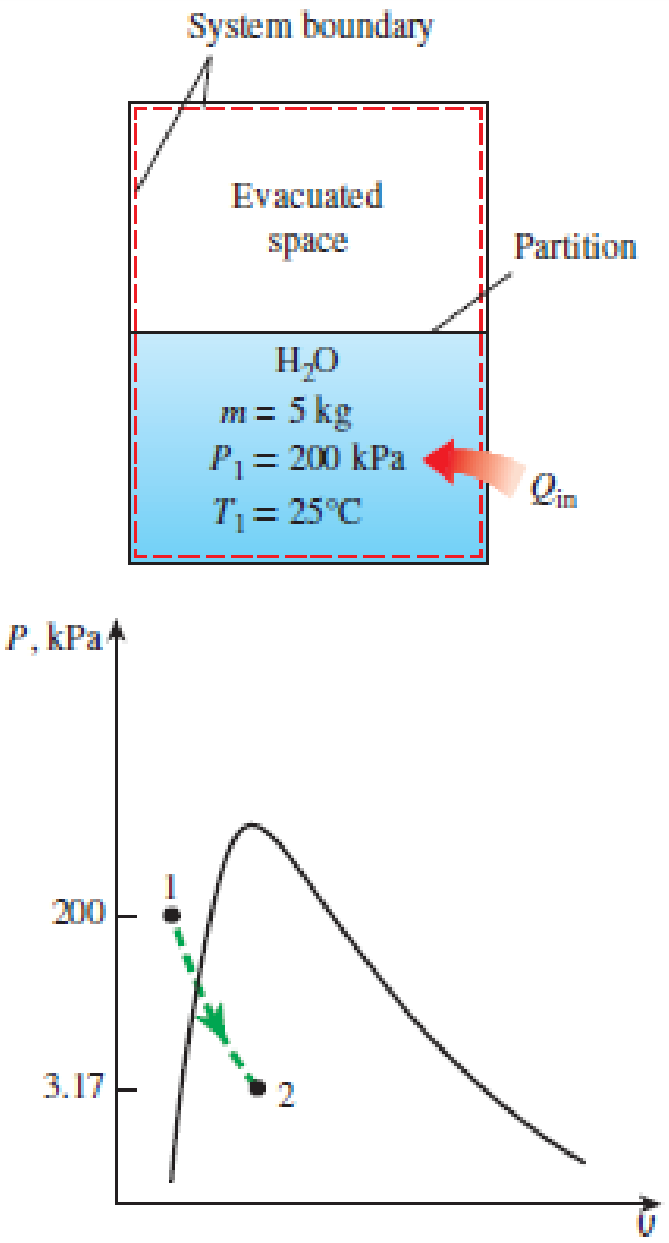


FIGURE 4–15
Schematic and P-v diagram for Example 4–6.

Notice that even though the water is expanding during this process, the system chosen involves fixed boundaries only (the dashed lines), and therefore the moving boundary work is zero (Fig. 4–16). Then $W = 0$ since the system does not involve any other forms of work. (Can you reach the same conclusion by choosing the water as our system?) Initially,

$$u_1 \cong u_{f\,@\,25°C} = 104.83 \text{ kJ/kg}$$

The quality at the final state is determined from the specific volume information:

$$x_2 = \frac{v_2 - v_f}{v_{fg}} = \frac{0.002 - 0.001}{43.34 - 0.001} = 2.3 \times 10^{-5}$$

Then

$$\begin{aligned} u_2 &= u_f + x_2 u_{fg} \\ &= 104.83 \text{ kJ/kg} + (2.3 \times 10^{-5})(2304.3 \text{ kJ/kg}) \\ &= 104.88 \text{ kJ/kg} \end{aligned}$$

Substituting yields

$$Q_{in} = (5 \text{ kg})[(104.88 - 104.83) \text{ kJ/kg}] = \mathbf{0.25\ kJ}$$

Discussion The positive sign indicates that the assumed direction is correct, and heat is transferred to the water.

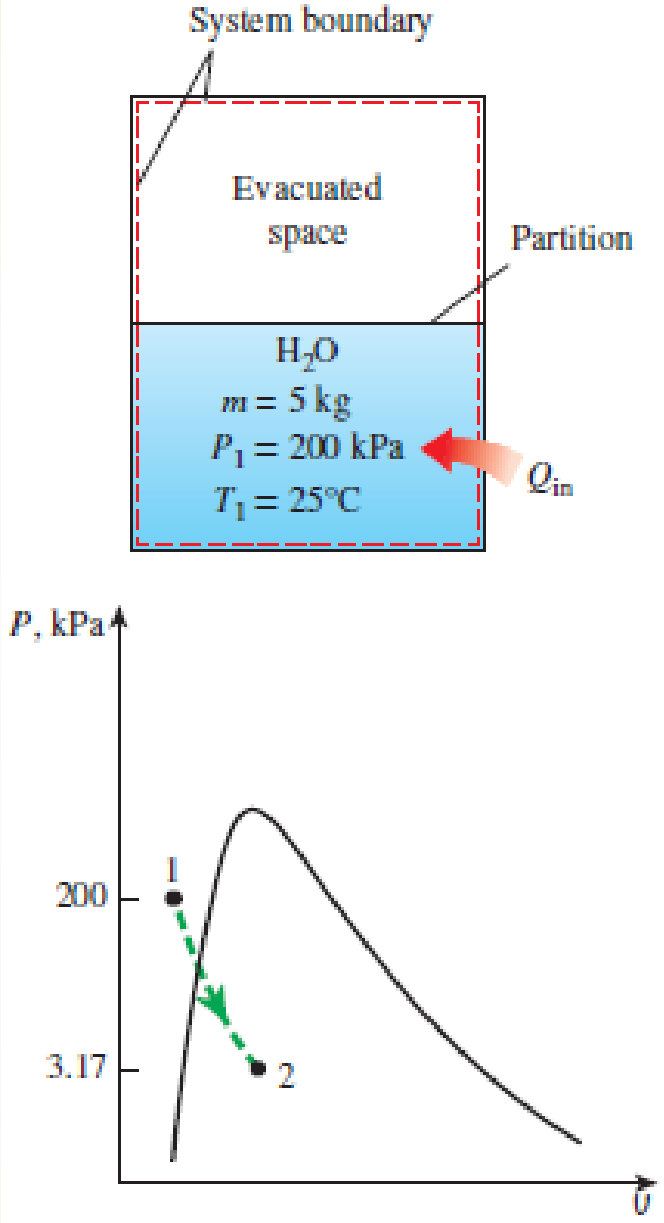


FIGURE 4–15
Schematic and P-v diagram for Example 4–6.

4.3 Specific heats

- **Specific heat** is defined as the energy required to raise the temperature of a unit mass of a substance by one degree.
- In general, this energy will depend on how the process is executed.
  1. With volume remain constant
  2. With pressure remain constant

- Specific heat at constant volume c_v
  - the energy required to raise the temperature of the unit mass of a substance by one degree as the volume is maintained constant.
- Specific heat at constant pressure c_p
  - the energy required to raise the temperature of the unit mass of a substance by one degree as the pressure is maintained constant.
- c_p is always greater than c_v since extra energy required for expansion work must also be supplied to the system

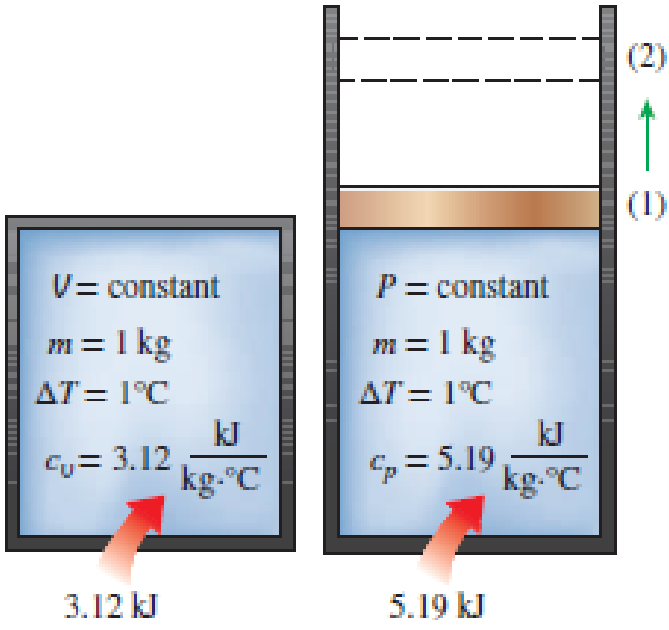


FIGURE 4–19
Constant-volume and constant-pressure specific heats c_v and c_p (values given are for helium gas).

4.3 Specific heats (2)

- Consider a fixed mass in a stationary closed system undergoing a **constant-volume process**.
- The conservation of energy principle $e_{in} - e_{out} = \Delta e_{system}$ for this process can be expressed as

$$\delta e_{in} - \delta e_{out} = du$$

- The left-hand side represents the net amount of energy transferred to the system.
- From the definition of c_v, this energy must be equal to $c_v dT$, where dT is the differential change in temperature. Thus, $c_v dT = du$ (at constant volume), or

$$c_v = \left(\frac{\partial u}{\partial T}\right)_v$$

the change in internal energy with temperature at constant volume

4.3 Specific heats (3)

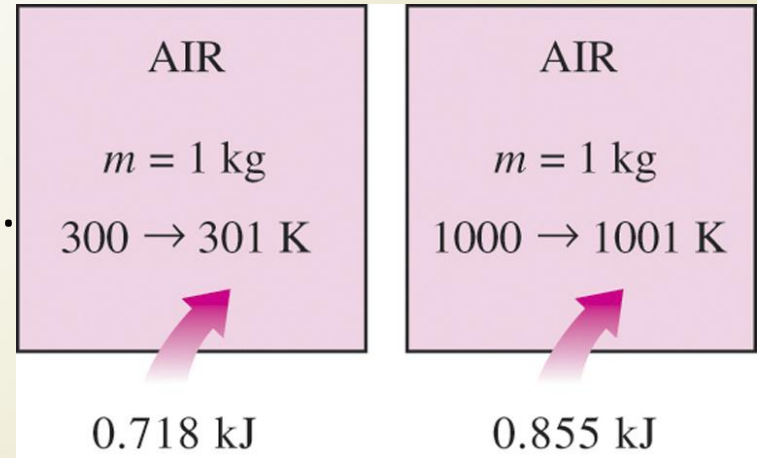
- Now, for the specific heat at constant pressure, c_p can be obtained by considering a constant-pressure expansion or compression process. It yields

$$\delta e_{in} - \delta e_{out} = du + dP_v = dh$$

$$c_p = \left(\frac{\partial h}{\partial T}\right)_p$$

the change in enthalpy with temperature at constant pressure

- c_v and c_p are expressed in terms of other properties; thus, they must be properties themselves and depend on the state (independent of the process).
  - i.e., the energy required to raise the temperature of a substance by one degree is different at different temperatures and pressures.
  - The difference is usually not very large.
- A common unit for specific heats is kJ/kg·°C or kJ/kg·K.
  - Note that these two units are identical since ΔT(°C) = ΔT(K)



4.4 Internal Energy, Enthalpy, and Specific Heats of Ideal Gases

- An ideal gas is a gas whose temperature, pressure, and specific volume are related by $Pv = RT$.
- It has been demonstrated mathematically and experimentally that for an ideal gas the internal energy is a function of the temperature only. That is,

$$u = u(T)$$

- Using the definition of enthalpy and the equation of state of an ideal gas, we have

$$\left.\begin{aligned} h &= u + Pv \\ Pv &= RT \end{aligned}\right\} h = u + RT$$

- Since R is constant and $u = u(T)$, it follows that the enthalpy of an ideal gas is also a function of temperature only: $h = h(T)$

4.4 Internal Energy, Enthalpy, and Specific Heats of Ideal Gases (2)

$$\Delta u = u_2 - u_1 \text{ (table)}$$

$$\Delta u = \int_1^2 c_v(T)\, dT$$

$$\Delta u \cong c_{v,\text{avg}}\, \Delta T$$

TABLE A–2

Ideal-gas specific heats of various common gases (*Concluded*)

(*c*) As a function of temperature

$$\bar{c}_p = a + bT + cT^2 + dT^3$$

(T in K, $\bar{c}_p$ in kJ/kmol · K)

Substance	Formula	a	b	c	d	Temperature range, K	% error Max.	% error Avg.
Nitrogen	N_2	28.90	-0.1571×10^{-2}	0.8081×10^{-5}	-2.873×10^{-9}	273–1800	0.59	0.34
Oxygen	O_2	25.48	1.520×10^{-2}	-0.7155×10^{-5}	1.312×10^{-9}	273–1800	1.19	0.28
Air	—	28.11	0.1967×10^{-2}	0.4802×10^{-5}	-1.966×10^{-9}	273–1800	0.72	0.33
Hydrogen	H_2	29.11	-0.1916×10^{-2}	0.4003×10^{-5}	-0.8704×10^{-9}	273–1800	1.01	0.26
Carbon								

Three ways of calculating Δu and Δh

- **By using the tabulated *u* and *h* data.**
  - This is the easiest and **most accurate** way when tables are readily available.
- **By using the c_v or c_p relations (Table A-2c) as a function of temperature and performing the integrations.**
  - This is very inconvenient for hand calculations but quite desirable for computerized calculations. The results obtained are **very accurate**.
- **By using average specific heats.**
  - This is very simple and certainly very convenient when property tables are not available. The results obtained are **reasonably accurate** if the temperature interval is not very large.

4.4 Internal Energy, Enthalpy, and Specific Heats of Ideal Gases (3)

- Since u and h depend only on temperature for an ideal gas, the specific heats c_v and c_p also depend on temperature only.
  - Therefore u, h, c_v and c_p of an ideal gas have fixed values regardless of the specific volume or pressure

$$u = u(T)$$
$$h = h(T)$$
$$c_v = c_v(T)$$
$$c_p = c_p(T)$$

For ideal gases, u, h, c_v and c_p vary with temperature only.

- Thus, for ideal gases, the partial derivatives can be replaced by ordinary derivatives.
- The differential changes in the internal energy and enthalpy of an ideal gas can be expressed as

$$du = c_v(T)\,dT; \qquad dh = c_p(T)\,dT$$

- The change in internal energy or enthalpy for an ideal gas during a process from state 1 to state 2 is determined by

$$\Delta u = u_2 - u_1 = \int_1^2 c_v(T)\,dT \quad (kJ/kg) \quad \text{and} \quad \Delta h = h_2 - h_1 = \int_1^2 c_p(T)\,dT \quad (kJ/kg)$$

4.4 Internal Energy, Enthalpy, and Specific Heats of Ideal Gases (4)

- The variation of specific heats with temperature is smooth and may be approximated as linear over small temperature intervals (a few hundred degrees or less).
- The specific heat functions then, can be replaced by the constant average specific heat values.
- Then the integration of these eqns. yields

$$u_2 - u_1 = c_{v,avg}\,(T_2 - T_1) \qquad \text{(kJ/kg)}$$

and

$$h_2 - h_1 = c_{p,avg}\,(T_2 - T_1) \qquad \text{(kJ/kg)}$$

- Δu and Δh relations are valid for all processes, not only control volume

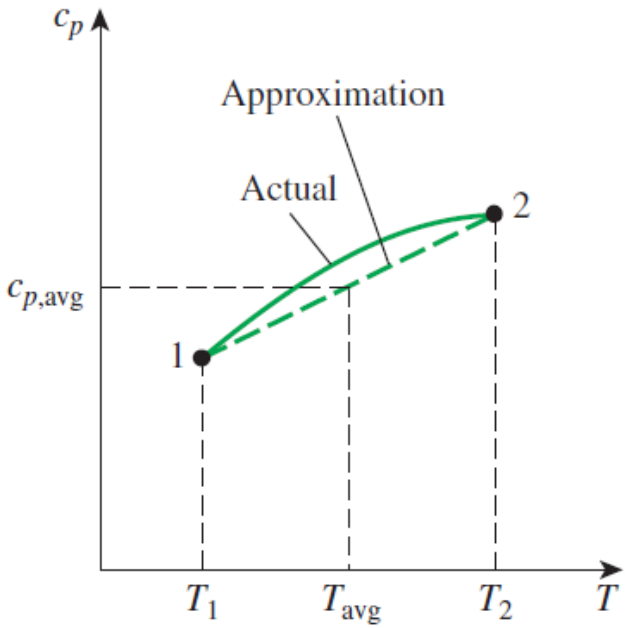


FIGURE 4–26
For small temperature intervals, the specific heats may be assumed to vary linearly with temperature.

4.4 Internal Energy, Enthalpy, and Specific Heats of Ideal Gases (4)

Specific Heat Relations of Ideal Gases

- A special relationship between c_v and c_p for ideal gases can be obtained by differentiating the relation $h = u + RT$, which yields $dh = du + RdT$
- Replacing dh by $c_p dT$ and du by $c_v dT$ and dividing the resulting expression by dT, we obtain

$$c_p = c_v + R \qquad \text{(kJ/kg.K)}$$

- Another ideal-gas property called the **specific heat ratio** k, is defined as

$$k = \frac{c_p}{c_v}$$

EXAMPLE 4–8 Heating of a Gas in a Tank by Stirring

An insulated rigid tank initially contains 1.5 lbm of helium at 80°F and 50 psia. A paddle wheel with a power rating of 0.02 hp is operated within the tank for 30 min. Determine (*a*) the final temperature and (*b*) the final pressure of the helium gas.

SOLUTION Helium gas in an insulated rigid tank is stirred by a paddle wheel. The final temperature and pressure of helium are to be determined.

Assumptions **1** Helium is an ideal gas since it is at a very high temperature relative to its critical-point value of −451°F. **2** Constant specific heats can be used for helium. **3** The system is stationary and thus the kinetic and potential energy changes are zero, $\Delta KE = \Delta PE = 0$ and $\Delta E = \Delta U$. **4** The volume of the tank is constant, and thus there is no boundary work. **5** The system is adiabatic and thus there is no heat transfer.

Analysis We take the contents of the tank as the *system* (Fig. 4–30). This is a *closed system* since no mass crosses the system boundary during the process. We observe that there is shaft work done on the system.

(*a*) The amount of paddle-wheel work done on the system is

$$W_{sh} = \dot{W}_{sh}\Delta t = (0.02\text{ hp})(0.5\text{ h})\left(\frac{2545\text{ Btu/h}}{1\text{ hp}}\right) = 25.45\text{ Btu}$$

Under the stated assumptions and observations, the energy balance on the system can be expressed as

$$\underbrace{E_{in} - E_{out}}_{\substack{\text{Net energy transfer}\\ \text{by heat, work, and mass}}} = \underbrace{\Delta E_{system}}_{\substack{\text{Change in internal, kinetic,}\\ \text{potential, etc., energies}}}$$

$$W_{sh,in} = \Delta U = m(u_2 - u_1) = mc_{v,avg}(T_2 - T_1)$$

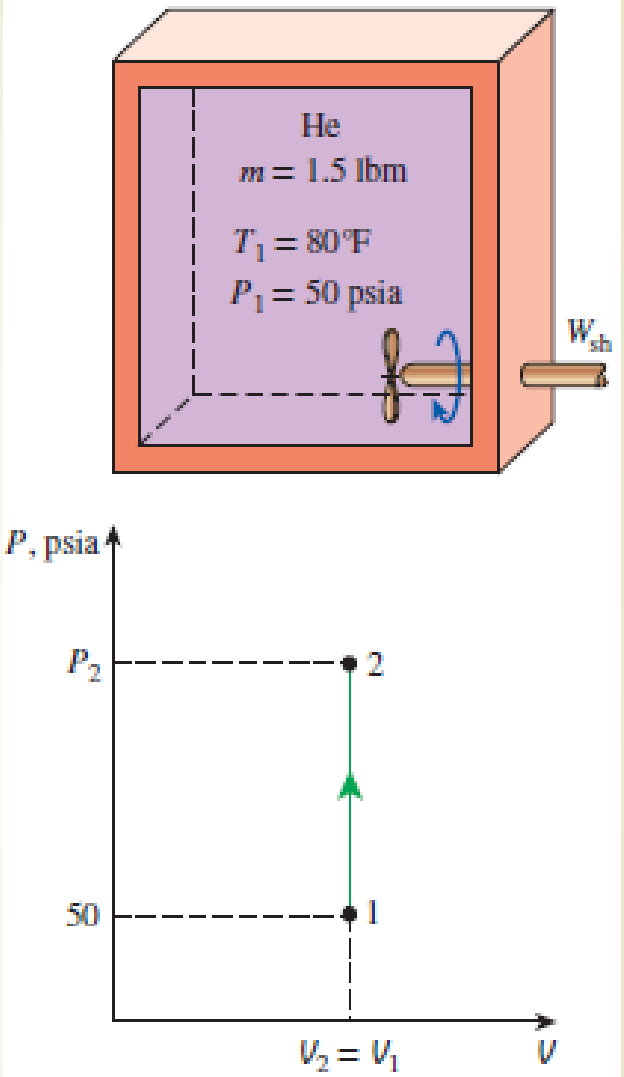


FIGURE 4–30
Schematic and *P*-*V* diagram for Example 4–8.

As we pointed out earlier, the ideal-gas specific heats of monatomic gases (helium being one of them) are constant. The c_v value of helium is determined from Table A–2E*a* to be $c_v = 0.753$ Btu/lbm·°F. Substituting this and other known quantities into the preceding equation, we obtain

$$25.45 \text{ Btu} = (1.5 \text{ lbm})(0.753 \text{ Btu/lbm}\cdot°\text{F})(T_2 - 80)°\text{F}$$
$$T_2 = \mathbf{102.5°F}$$

(*b*) The final pressure is determined from the ideal-gas relation

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

where V_1 and V_2 are identical and cancel out. Then the final pressure becomes

$$\frac{50 \text{ psia}}{(80 + 460)\text{R}} = \frac{P_2}{(102.5 + 460)R}$$
$$P_2 = \mathbf{52.1 \text{ psia}}$$

Discussion Note that the pressure in the ideal-gas relation is always the absolute pressure.

Redo for the following:

Gas = He

m = 0.7 kg

T_1 = 27°C

P_1 = 350 kPa

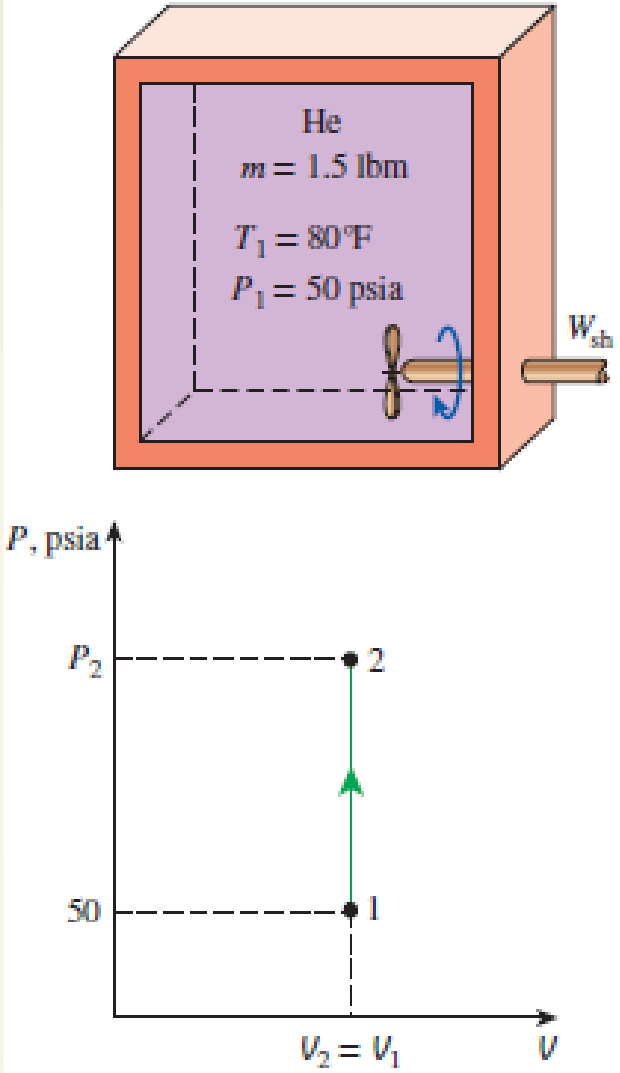


FIGURE 4–30
Schematic and *P*-*V* diagram for Example 4–8.

EXAMPLE 4–9 Heating of a Gas by a Resistance Heater

A piston–cylinder device initially contains 0.5 m^3 of nitrogen gas at 400 kPa and 27°C. An electric heater within the device is turned on and is allowed to pass a current of 2 A for 5 min from a 120-V source. Nitrogen expands at constant pressure, and a heat loss of 2800 J occurs during the process. Determine the final temperature of nitrogen.

SOLUTION Nitrogen gas in a piston–cylinder device is heated by an electric resistance heater. Nitrogen expands at constant pressure while some heat is lost. The final temperature of nitrogen is to be determined.

Assumptions **1** Nitrogen is an ideal gas since it is at a high temperature and low pressure relative to its critical-point values of −147°C and 3.39 MPa. **2** The system is stationary, and thus the kinetic and potential energy changes are zero, $\Delta KE = \Delta PE = 0$ and $\Delta E = \Delta U$. **3** The pressure remains constant during the process and thus $P_2 = P_1$. **4** Nitrogen has constant specific heats at room temperature.

Analysis We take the contents of the cylinder as the *system* (Fig. 4–31). This is a *closed system* since no mass crosses the system boundary during the process. We observe that a piston–cylinder device typically involves a moving boundary and thus boundary work, W_b. Also, heat is lost from the system and electrical work W_e is done on the system.

First, let us determine the electrical work done on the nitrogen:

$$W_e = \mathbf{V}I\,\Delta t = (120\text{ V})(2\text{ A})(5\times 60\text{ s})\left(\frac{1\text{ kJ/s}}{1000\text{ VA}}\right) = 72\text{ kJ}$$

The mass of nitrogen is determined from the ideal-gas relation:

$$m = \frac{P_1V_1}{RT_1} = \frac{(400\text{ kPa})(0.5\text{ m}^3)}{(0.297\text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K})(300\text{ K})} = 2.245\text{ kg}$$

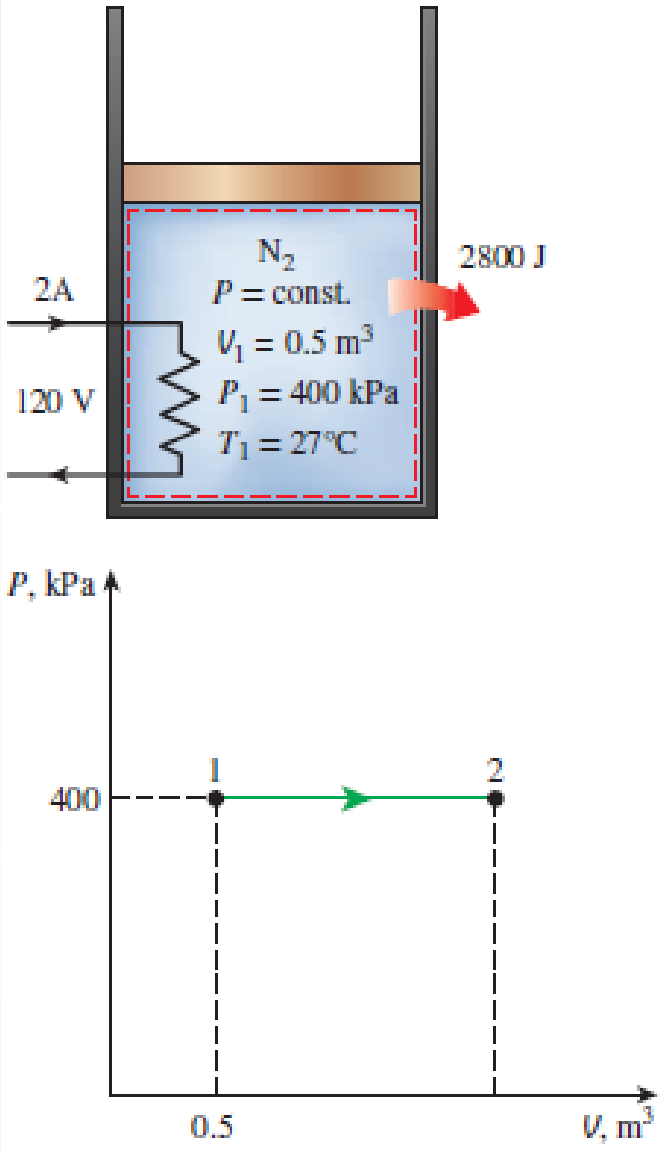


FIGURE 4–31
Schematic and *P*-*V* diagram for Example 4–9.

Under the stated assumptions and observations, the energy balance on the system can be expressed as

$$\underbrace{E_{in} - E_{out}}_{\substack{\text{Net energy transfer}\\ \text{by heat, work, and mass}}} = \underbrace{\Delta E_{system}}_{\substack{\text{Change in internal, kinetic,}\\ \text{potential, etc., energies}}}$$

$$W_{e,in} - Q_{out} - W_{b,out} = \Delta U$$

$$W_{e,in} - Q_{out} = \Delta H = m(h_2 - h_1) = mc_p(T_2 - T_1)$$

since $\Delta U + W_b = \Delta H$ for a closed system undergoing a quasi-equilibrium expansion or compression process at constant pressure. From Table A–2*a*, $c_p = 1.039$ kJ/kg·K for nitrogen at room temperature. The only unknown quantity in the previous equation is T_2, and it is found to be

$$72 \text{ kJ} - 2.8 \text{ kJ} = (2.245 \text{ kg})(1.039 \text{ kJ/kg·K})(T_2 - 27)°\text{C}$$

$$T_2 = \mathbf{56.7°C}$$

Discussion Note that we could also solve this problem by determining the boundary work and the internal energy change rather than the enthalpy change.

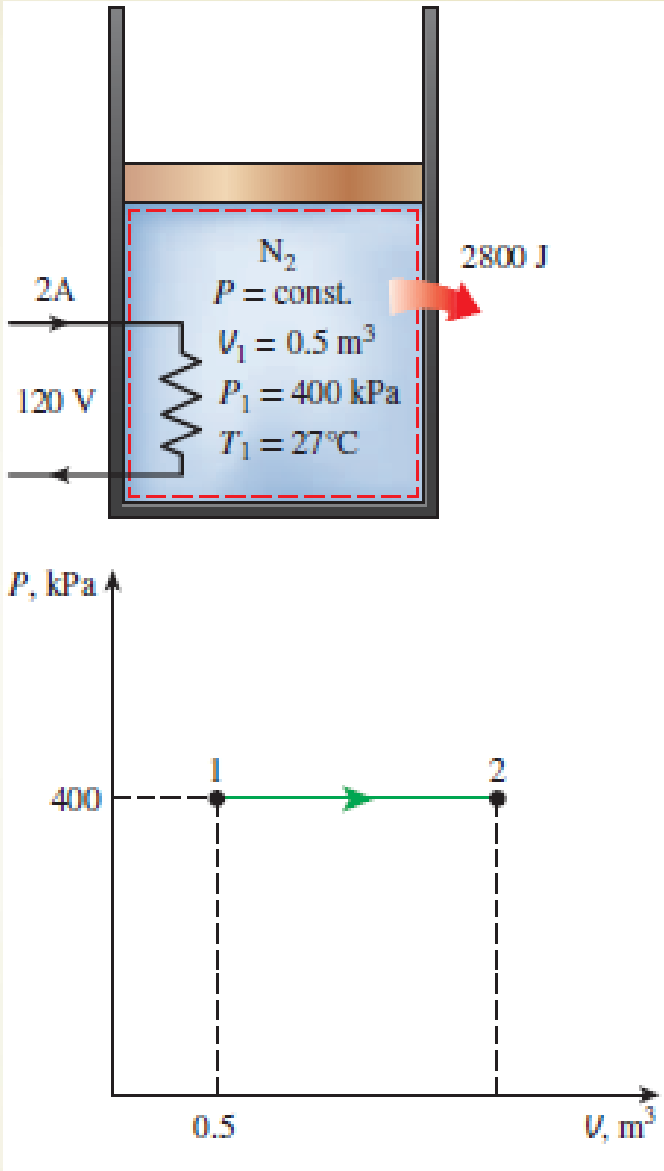


FIGURE 4–31
Schematic and *P*-*V* diagram for Example 4–9.

EXAMPLE 4–10 Heating of a Gas at Constant Pressure

A piston–cylinder device initially contains air at 150 kPa and 27°C. At this state, the piston is resting on a pair of stops, as shown in Fig. 4–32, and the enclosed volume is 400 L. The mass of the piston is such that a 350-kPa pressure is required to move it. The air is now heated until its volume has doubled. Determine (*a*) the final temperature, (*b*) the work done by the air, and (*c*) the total heat transferred to the air.

SOLUTION Air in a piston–cylinder device with a set of stops is heated until its volume is doubled. The final temperature, work done, and the total heat transfer are to be determined.

Assumptions **1** Air is an ideal gas since it is at a high temperature and low pressure relative to its critical-point values. **2** The system is stationary and thus the kinetic and potential energy changes are zero, $\Delta KE = \Delta PE = 0$ and $\Delta E = \Delta U$. **3** The volume remains constant until the piston starts moving, and the pressure remains constant afterwards. **4** There are no electrical, shaft, or other forms of work involved.

Analysis We take the contents of the cylinder as the *system* (Fig. 4–32). This is a *closed system* since no mass crosses the system boundary during the process. We observe that a piston–cylinder device typically involves a moving boundary and thus boundary work, W_b. Also, the boundary work is done by the system, and heat is transferred to the system.

(*a*) The final temperature can be determined easily by using the ideal-gas relation between states 1 and 3 in the following form:

$$\frac{P_1V_1}{T_1} = \frac{P_3V_3}{T_3} \longrightarrow \frac{(150\text{ kPa})(V_1)}{300\text{ K}} = \frac{(350\text{ kPa})(2V_1)}{T_3}$$

$$T_3 = \mathbf{1400\ K}$$

(*b*) The work done could be determined by integration, but for this case it is much easier to find it from the area under the process curve on a *P-V* diagram, shown in Fig. 4–32:

$$A = (V_2 - V_1)P_2 = (0.4\text{ m}^3)(350\text{ kPa}) = 140\text{ m}^3\cdot\text{kPa}$$

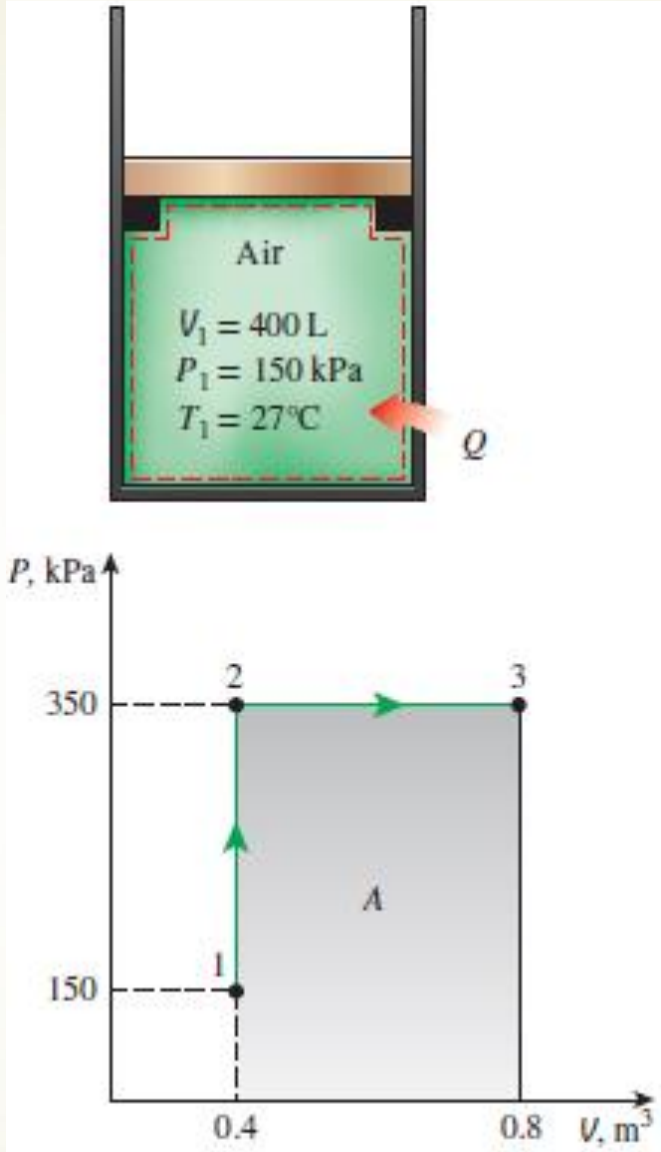


FIGURE 4–32
Schematic and *P-V* diagram for Example 4–10.

Therefore,

$$W_{13} = \mathbf{140\ kJ}$$

The work is done by the system (to raise the piston and to push the atmospheric air out of the way), and thus it is work output.

(*c*) Under the stated assumptions and observations, the energy balance on the system between the initial and final states (process 1–3) can be expressed as

$$\underbrace{E_{in} - E_{out}}_{\substack{\text{Net energy transfer}\\ \text{by heat, work, and mass}}} = \underbrace{\Delta E_{system}}_{\substack{\text{Change in internal, kinetic,}\\ \text{potential, etc., energies}}}$$

$$Q_{in} - W_{b,out} = \Delta U = m(u_3 - u_1)$$

The mass of the system can be determined from the ideal-gas relation:

$$m = \frac{P_1 V_1}{RT_1} = \frac{(150\ \text{kPa})(0.4\ \text{m}^3)}{(0.287\ \text{kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K})(300\ \text{K})} = 0.697\ \text{kg}$$

The internal energies are determined from the air table (Table A–17) to be

$$u_1 = u_{@\,300\ \text{K}} = 214.07\ \text{kJ/kg}$$

$$u_3 = u_{@\,1400\ \text{K}} = 1113.52\ \text{kJ/kg}$$

Thus,

$$Q_{in} - 140\ \text{kJ} = (0.697\ \text{kg})[(1113.52 - 214.07)\ \text{kJ/kg}]$$

$$Q_{in} = \mathbf{767\ kJ}$$

Discussion The positive sign verifies that heat is transferred to the system.

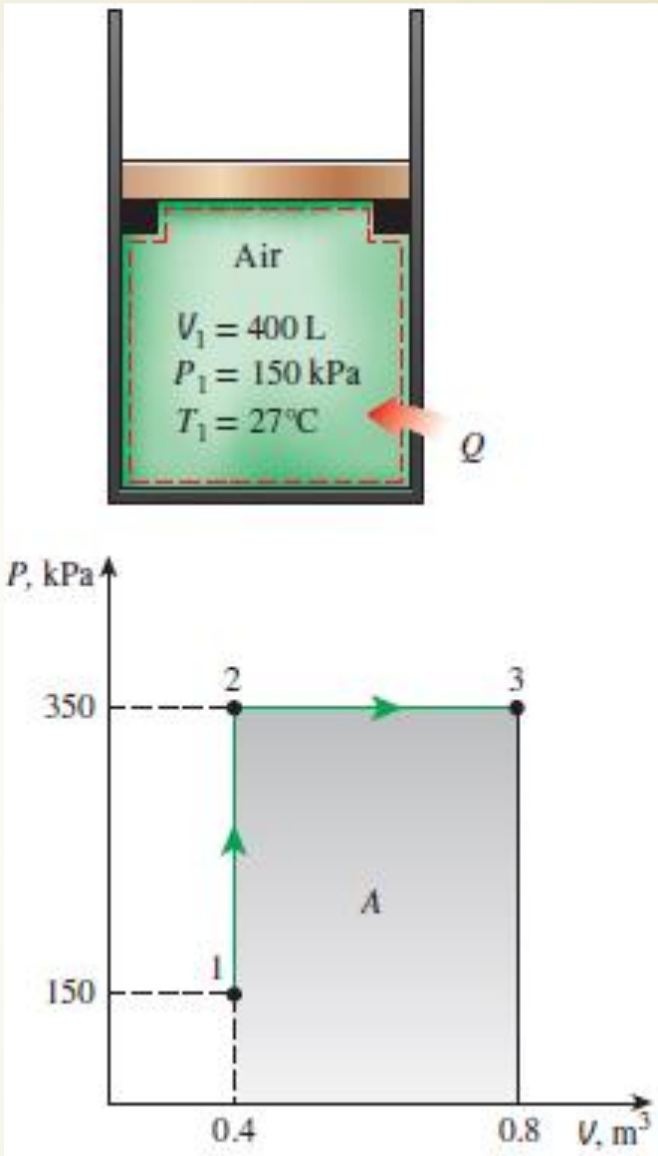
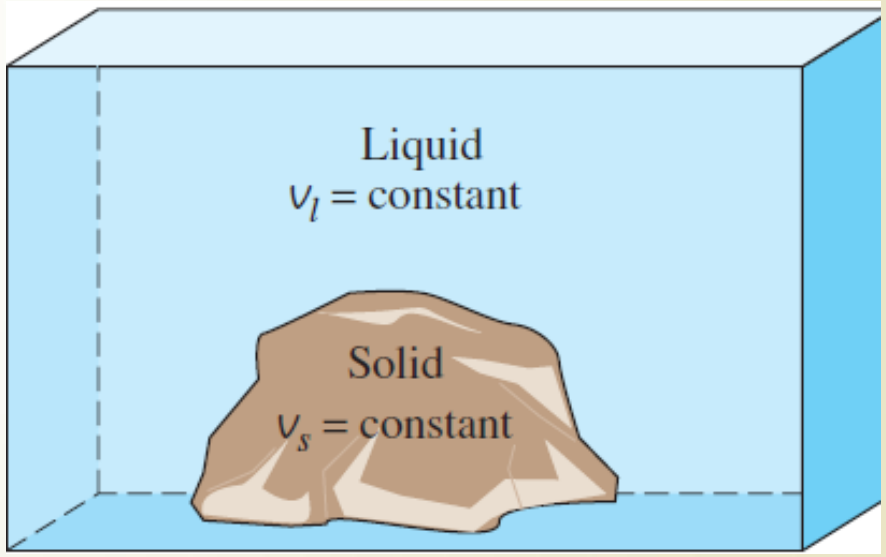


FIGURE 4–32
Schematic and *P-V* diagram for Example 4–10.

4.5 Internal Energy, Enthalpy, and Specific Heats of Solids and Liquids

- A substance whose specific volume (or density) is constant is called an **incompressible substance**.
- The specific volumes of solids and liquids essentially remain constant during a process and can be approximated as incompressible substance.
- Constant-volume and constant-pressure specific heats are identical for incompressible substances.
- Therefore, for solids and liquids, the subscript on c_v and c_p can be dropped, and both specific heats can be represented by a single symbol c.
- That is, $\boldsymbol{c_p = c_v = c}$



4.5 Internal Energy, Enthalpy, and Specific Heats of Solids and Liquids (2)

Internal Energy Changes

- Like those of ideal gases, the specific heats of incompressible substances depend on temperature only.

$$du = c_v\, dT = c(T)dT$$

- The change in internal energy between states 1 and 2 is then obtained by integration:

$$\Delta u = u_2 - u_1 = \int_1^2 c(T)dT \qquad (kJ/kg)$$

- For small temperature intervals, a *c* value at the average temperature can be used and treated as a constant, yielding

$$\Delta u \cong c_{avg}(T_2 - T_1) \qquad (kJ/kg)$$

4.5 Internal Energy, Enthalpy, and Specific Heats of Solids and Liquids (2)

Enthalpy Changes

- Using the definition of enthalpy $h = u + Pv$ and noting that v = constant, the differential form of the enthalpy change of incompressible substances can be determined by differentiation,

$$dh = du + v\,dP + P\,\cancelto{0}{dv} = du + v\,dP$$

- Integrating,

$$\Delta h = \Delta u + v\,\Delta P \cong c_{avg}\,\Delta T + v\,\Delta P \qquad \text{(kJ/kg)}$$

For solids, the term $v\,\Delta P$ is insignificant and thus $\Delta h \cong \Delta u \cong c_{avg}\,\Delta T$.

- For liquids, two special cases are commonly encountered:
  - **Constant-pressure processes**, as in heaters $(\Delta P = 0)$: $\Delta h = \Delta u \cong c_{avg}\Delta T$
  - **Constant-temperature processes**, as in pumps $(\Delta T = 0)$: $\Delta h = v\,\Delta P$

EXAMPLE 4–12 Cooling of an Iron Block by Water

A 50-kg iron block at 80°C is dropped into an insulated tank that contains 0.5 m^3 of liquid water at 25°C. Determine the temperature when thermal equilibrium is reached.

SOLUTION An iron block is dropped into water in an insulated tank. The final temperature when thermal equilibrium is reached is to be determined.

Assumptions **1** Both water and the iron block are incompressible substances. **2** Constant specific heats at room temperature can be used for water and the iron. **3** The system is stationary, and thus the kinetic and potential energy changes are zero, $\Delta KE = \Delta PE = 0$ and $\Delta E = \Delta U$. **4** There are no electrical, shaft, or other forms of work involved. **5** The system is well insulated and thus there is no heat transfer.

Analysis We take the entire contents of the tank as the *system* (Fig. 4–35). This is a *closed system* since no mass crosses the system boundary during the process. We observe that the volume of a rigid tank is constant, and thus there is no boundary work. The energy balance on the system can be expressed as

$$\underbrace{E_{in} - E_{out}}_{\substack{\text{Net energy transfer}\\ \text{by heat, work, and mass}}} = \underbrace{\Delta E_{system}}_{\substack{\text{Change in internal, kinetic,}\\ \text{potential, etc., energies}}}$$

$$0 = \Delta U$$

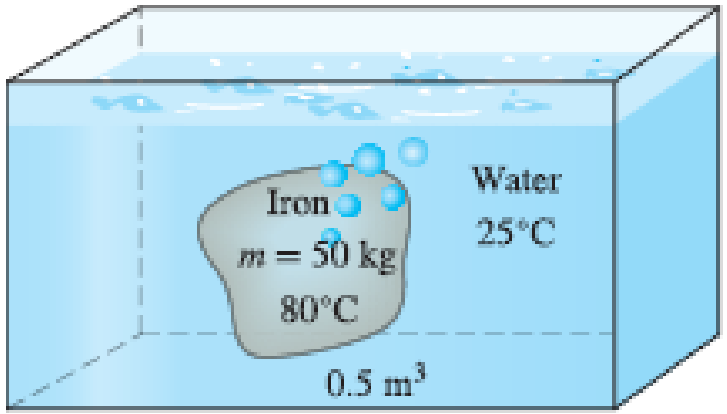


FIGURE 4–35
Schematic for Example 4–12.

The total internal energy U is an extensive property, and therefore it can be expressed as the sum of the internal energies of the parts of the system. Then the total internal energy change of the system becomes

$$\Delta U_{sys} = \Delta U_{iron} + \Delta U_{water} = 0$$
$$[mc(T_2 - T_1)]_{iron} + [mc(T_2 - T_1)]_{water} = 0$$

The specific volume of liquid water at or about room temperature can be taken to be 0.001 m³/kg. Then the mass of the water is

$$m_{water} = \frac{V}{v} = \frac{0.5\ \text{m}^3}{0.001\ \text{m}^3/\text{kg}} = 500\ \text{kg}$$

The specific heats of iron and liquid water are determined from Table A–3 to be $c_{iron} = 0.45$ kJ/kg·°C and $c_{water} = 4.18$ kJ/kg·°C. Substituting these values into the energy equation, we obtain

$$(50\ \text{kg})(0.45\ \text{kJ/kg}\cdot°\text{C})(T_2 - 80°\text{C}) + (500\ \text{kg})(4.18\ \text{kJ/kg}\cdot°\text{C})(T_2 - 25°\text{C}) = 0$$

$$T_2 = \mathbf{25.6°C}$$

Therefore, when thermal equilibrium is established, both the water and iron will be at 25.6°C.

Discussion The small rise in water temperature is due to its large mass and large specific heat.

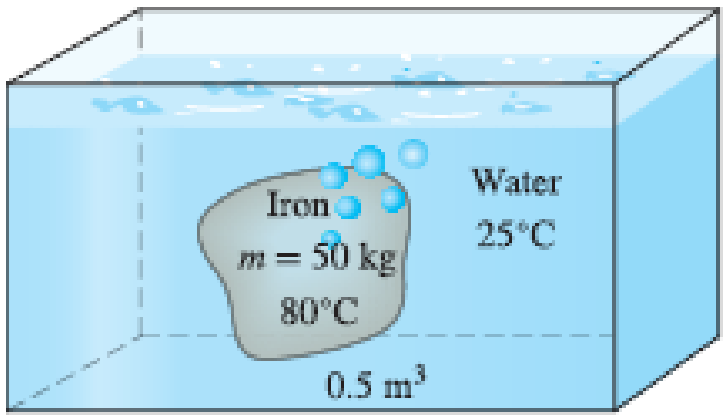


FIGURE 4–35
Schematic for Example 4–12.

EXAMPLE 4–13 Cooling of Carbon Steel Balls in Air

Carbon steel balls ($\rho = 7833$ kg/m^3 and $c_p = 0.465$ kJ/kg·°C) 8 mm in diameter are annealed by heating them first to 900°C in a furnace, and then allowing them to cool slowly to 100°C in ambient air at 35°C, as shown in Fig. 4–36. If 2500 balls are to be annealed per hour, determine the total rate of heat transfer from the balls to the ambient air.

SOLUTION Carbon steel balls are to be annealed at a rate of 2500/h by heating them first and then allowing them to cool slowly in ambient air at a specified rate. The total rate of heat transfer from the balls to the ambient air is to be determined.

Assumptions **1** The thermal properties of the balls are constant. **2** There are no changes in kinetic and potential energies. **3** The balls are at a uniform temperature at the end of the process.

Properties The density and specific heat of the balls are given to be $\rho = 7833$ kg/m^3 and $c_p = 0.465$ kJ/kg·°C.

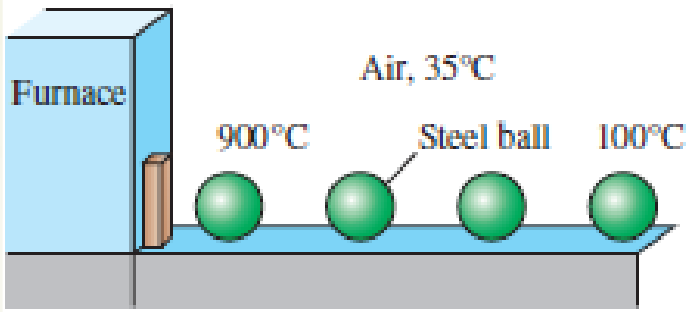


FIGURE 4–36
Schematic for Example 4–13.

Analysis We take a single ball as the system. The energy balance for this closed system can be expressed as

$$\underbrace{E_{in} - E_{out}}_{\substack{\text{Net energy transfer}\\ \text{by heat, work, and mass}}} = \underbrace{\Delta E_{system}}_{\substack{\text{Change in internal, kinetic,}\\ \text{potential, etc., energies}}}$$

$$-Q_{out} = \Delta U_{ball} = m(u_2 - u_1)$$

$$Q_{out} = mc(T_1 - T_2)$$

The amount of heat transfer from a single ball is

$$m = \rho V = \rho \frac{\pi D^3}{6} = (7833 \text{ kg/m}^3)\frac{\pi(0.008 \text{ m})^3}{6} = 0.00210 \text{ kg}$$

$$Q_{out} = mc(T_1 - T_2) = (0.00210 \text{ kg})(0.465 \text{ kJ/kg}\cdot{}^\circ\text{C})(900 - 100)^\circ\text{C}$$
$$= 0.781 \text{ kJ (per ball)}$$

Then the total rate of heat transfer from the balls to the ambient air becomes

$$\dot{Q}_{out} = \dot{n}_{ball} Q_{out} = (2500 \text{ ball/h})(0.781 \text{ kJ/ball}) = 1953 \text{ kJ/h} = \mathbf{542\ W}$$

Discussion For solids and liquids, constant-pressure and constant-volume specific heats are identical and can be represented by a single symbol c. However, it is customary to use the symbol c_p for the specific heat of incompressible substances.

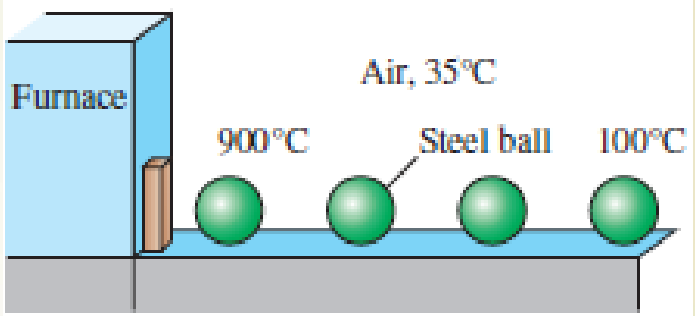


FIGURE 4–36
Schematic for Example 4–13.

$$V = \frac{4}{3}\pi r^3$$

Exercise 1

4–7 A piston–cylinder device initially contains 0.07 m^3 of nitrogen gas at 130 kPa and 180°C. The nitrogen is now expanded to a pressure of 80 kPa polytropically with a polytropic exponent whose value is equal to the specific heat ratio (called *isentropic expansion*). Determine the final temperature and the boundary work done during this process.

Exercise 2

4–18 A piston–cylinder device initially contains 0.4 kg of nitrogen gas at 160 kPa and 140°C. The nitrogen is now expanded isothermally to a pressure of 100 kPa. Determine the boundary work done during this process. *Answer:* 23.0 kJ

Exercise 3

4–32 A fixed mass of saturated water vapor at 400 kPa is isothermally cooled until it is a saturated liquid. Calculate the amount of heat rejected during this process, in kJ/kg.

Exercise 4

4–37 A piston–cylinder device initially contains 0.8 m^3 of saturated water vapor at 250 kPa. At this state, the piston is resting on a set of stops, and the mass of the piston is such that a pressure of 300 kPa is required to move it. Heat is now slowly transferred to the steam until the volume doubles. Show the process on a P-v diagram with respect to saturation lines and determine (*a*) the final temperature, (*b*) the work done during this process, and (*c*) the total heat transfer.

Answers: (*a*) 662°C, (*b*) 240 kJ, (*c*) 1213 kJ

Exercise 5

4–63 A 4-m × 5-m × 6-m room is to be heated by a baseboard resistance heater. It is desired that the resistance heater be able to raise the air temperature in the room from 5 to 25°C within 11 min. Assuming no heat losses from the room and an atmospheric pressure of 100 kPa, determine the required power of the resistance heater. Assume constant specific heats at room temperature. *Answer:* 3.28 kW

Exercise 6

4–76 Air is contained in a cylinder device fitted with a piston-cylinder. The piston initially rests on a set of stops, and a pressure of 200 kPa is required to move the piston. Initially, the air is at 100 kPa and 23°C and occupies a volume of 0.25 m^3. Determine the amount of heat transferred to the air, in kJ, while increasing the temperature to 700 K. Assume air has constant specific heats evaluated at 300 K. *Answer:* 94.5 kJ

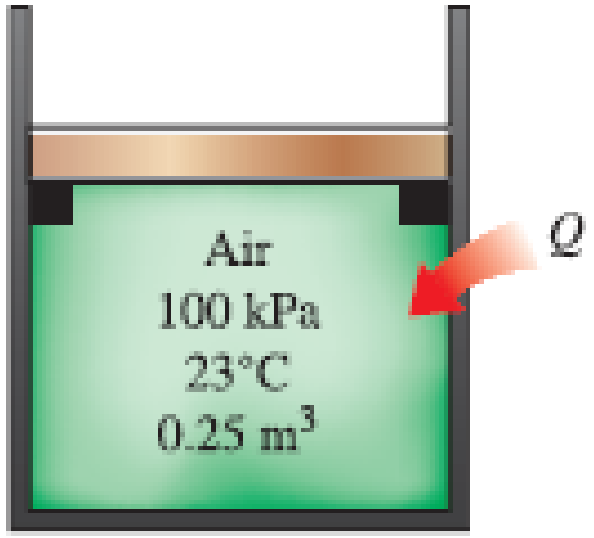


FIGURE P4–76

Thank You